AP CALCULUS AB

Unit 1 • Lesson 1.5 practice

Lesson 1.5 • Determining Limits Using Algebraic Properties of Limits

4 multiple-choice questions and 1 free-response question

Question 1

$\lim_{x \to 2} \left(3x^2 - x\right)$ is

(A) 10

(B) 12

(C) 14

(D) 8

Question 2

If $\lim_{x \to c} f(x) = 4$ and $\lim_{x \to c} g(x) = -1$, then $\lim_{x \to c} \left(f(x)g(x) + 3\right)$ is

(A) 7

(B) -1

(C) 3

(D) -4

Question 3

If $\lim_{x \to c} f(x) = 6$ and $\lim_{x \to c} g(x) = 2$, then $\lim_{x \to c} \dfrac{f(x)}{g(x)}$ is

(A) 12

(B) 4

(C) 3

(D) $\dfrac{1}{3}$

Question 4

$\lim_{x \to \pi} \cos x$ is

(A) 0

(B) 1

(C) π

(D) -1

Question 5 (Free response)

$\lim_{x \to 1} f(x) = 3$, $\lim_{x \to 1} g(x) = -2$ and $h(x) = x^2 + 1$. Find each limit, showing which properties you use.

(a) $\lim_{x \to 1} \left(2f(x) - g(x)\right)$

(b) $\lim_{x \to 1} f(x)h(x)$

(c) $\lim_{x \to 1} \sqrt{f(x) + 6}$

(d) $\lim_{x \to 1} \left(g(x)\right)^3$